5th–8th Grade Math Complete

A Comprehensive Course for Middle School Students

Target Audience: Grades 5–8

Lazard Legacy Math Academy

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"Dreams can come true. Better late than never."

Welcome to the Lazard Legacy Math Academy!

This course book is designed to help you master the essential math skills needed for success in middle school and beyond. Whether you are preparing for the LEAP 2025 test, building your confidence, or strengthening your foundation — this book has you covered.

What You Will Learn:

• How to add, subtract, multiply, and divide fractions with confidence

• How to work with decimals and percentages in real-world situations

• How to solve ratios and proportions

• How to handle positive and negative integers

• How to write and solve algebraic expressions and equations

• A complete LEAP 2025 practice test with answer key

How to Use This Book:

1. Read each lesson carefully and study the rules in the boxes.
2. Follow along with every worked example step by step.
3. Try the practice problems on your own BEFORE checking answers.
4. Use the LEAP practice test to check your overall understanding.

"The only way to learn mathematics is to DO mathematics." — Paul Halmos

Table of Contents

Section 1: Fractions

Add, Subtract, Multiply, and Divide

Fractions are one of the most important math concepts you will use in middle school and beyond. A fraction represents a part of a whole. The top number (numerator) tells how many parts you have, and the bottom number (denominator) tells how many equal parts the whole is divided into.

Key Vocabulary:

• **Numerator:** The top number in a fraction — how many parts you have

• **Denominator:** The bottom number — how many equal parts in the whole

• **LCD (Least Common Denominator):** The smallest number that both denominators divide into evenly

• **GCF (Greatest Common Factor):** The largest number that divides evenly into both numbers — used to simplify

• **Reciprocal:** A fraction flipped upside down (3/4 → 4/3) — used in division

• **Simplify:** Reduce a fraction to lowest terms by dividing by the GCF

Types of Fractions:

• **Proper:** Numerator < Denominator (example: 3/4) — less than 1 whole

• **Improper:** Numerator ≥ Denominator (example: 7/4) — 1 or more wholes

• **Mixed Number:** Whole + fraction (example: 1 3/4)

1.1 Adding Fractions

Rule: To add fractions with the same denominator, add the numerators and keep the denominator the same. For fractions with different denominators, first find the Least Common Denominator (LCD), rewrite each fraction with the LCD, then add.

Step-by-Step Process:

Step 1: Look at the denominators. Are they the same or different?

Step 2: If different, find the LCD (list multiples of each denominator).

Step 3: Rewrite each fraction as an equivalent fraction with the LCD.

Step 4: Add the numerators. Keep the common denominator.

Step 5: Simplify the result (divide by GCF). Convert improper to mixed if needed.

Worked Examples:

Example 1: 1/4 + 2/4

Same denominator (4)! Just add numerators:
1 + 2 = 3, keep denominator 4
Answer: 3/4

Example 2: 2/5 + 1/5

Same denominator (5)!
2 + 1 = 3, keep denominator 5
Answer: 3/5

Example 3: 1/3 + 1/6

Different denominators! Find LCD of 3 and 6:
Multiples of 3: 3, 6, 9... Multiples of 6: 6, 12...
LCD = 6
Rewrite: 1/3 = 2/6 (multiply top and bottom by 2)
Add: 2/6 + 1/6 = 3/6
Simplify: 3/6 = 1/2
Answer: 1/2

Example 4: 3/8 + 1/4

LCD of 8 and 4 = 8
Rewrite 1/4 = 2/8
3/8 + 2/8 = 5/8
Answer: 5/8

Example 5: 2/3 + 1/4

LCD of 3 and 4 = 12
2/3 = 8/12 (multiply by 4)
1/4 = 3/12 (multiply by 3)
8/12 + 3/12 = 11/12
Answer: 11/12

1.1 Adding Fractions — More Examples

Example 6: 5/6 + 1/3

LCD of 6 and 3 = 6
1/3 = 2/6
5/6 + 2/6 = 7/6
Improper! Convert: 7 ÷ 6 = 1 remainder 1 → 1 1/6
Answer: 1 1/6

Example 7: 1/2 + 3/8

LCD of 2 and 8 = 8
1/2 = 4/8
4/8 + 3/8 = 7/8
Answer: 7/8

Example 8: 4/9 + 2/3

LCD of 9 and 3 = 9
2/3 = 6/9
4/9 + 6/9 = 10/9 = 1 1/9
Answer: 1 1/9

Example 9: 3/5 + 2/10

LCD of 5 and 10 = 10
3/5 = 6/10
6/10 + 2/10 = 8/10 = 4/5
Answer: 4/5

Example 10: 7/12 + 1/4

LCD of 12 and 4 = 12
1/4 = 3/12
7/12 + 3/12 = 10/12 = 5/6

Answer: 5/6

1.2 Subtracting Fractions

Rule: To subtract fractions with the same denominator, subtract the numerators and keep the denominator. For different denominators, find the LCD first, rewrite each fraction, then subtract. The process is the same as addition — just subtract instead of add.

Step-by-Step Process:

Step 1: Check denominators — same or different?

Step 2: If different, find the LCD.

Step 3: Rewrite fractions with the LCD.

Step 4: Subtract the numerators. Keep the denominator.

Step 5: Simplify if possible.

Worked Examples:

Example 1: 3/4 – 1/4

Same denominator!
3 – 1 = 2, keep denominator 4
2/4 → simplify: GCF = 2 → 1/2
Answer: 1/2

Example 2: 5/6 – 1/6

Same denominator: 5 – 1 = 4
4/6 → simplify: GCF = 2 → 2/3
Answer: 2/3

Example 3: 7/8 – 3/8

Same denominator: 7 – 3 = 4
4/8 → simplify: 1/2
Answer: 1/2

Example 4: 2/3 – 1/4

LCD of 3 and 4 = 12
2/3 = 8/12, 1/4 = 3/12
8/12 – 3/12 = 5/12
Answer: 5/12

Example 5: 5/6 – 1/3

LCD of 6 and 3 = 6
1/3 = 2/6
5/6 – 2/6 = 3/6 = 1/2
Answer: 1/2

1.2 Subtracting Fractions — More Examples

Example 6: 3/4 – 2/5

LCD of 4 and 5 = 20
3/4 = 15/20, 2/5 = 8/20
15/20 – 8/20 = 7/20
Answer: 7/20

Example 7: 7/10 – 1/5

LCD of 10 and 5 = 10
1/5 = 2/10
7/10 – 2/10 = 5/10 = 1/2
Answer: 1/2

Example 8: 11/12 – 3/4

LCD of 12 and 4 = 12
3/4 = 9/12
11/12 – 9/12 = 2/12 = 1/6
Answer: 1/6

Example 9: 4/5 – 1/3

LCD of 5 and 3 = 15
4/5 = 12/15, 1/3 = 5/15
12/15 – 5/15 = 7/15
Answer: 7/15

Example 10: 9/10 – 2/5

LCD of 10 and 5 = 10
2/5 = 4/10
9/10 – 4/10 = 5/10 = 1/2
Answer: 1/2

1.3 Multiplying Fractions

Rule: To multiply fractions, multiply the numerators together and multiply the denominators together. Simplify the result. You do NOT need a common denominator for multiplication!

Cross-Canceling (Shortcut): Before multiplying, look for common factors between any numerator and any denominator. Divide them out first — this gives you smaller numbers to multiply!

Step-by-Step:

Step 1: Look for opportunities to cross-cancel.

Step 2: Multiply numerators (top × top).

Step 3: Multiply denominators (bottom × bottom).

Step 4: Simplify if you didn't cross-cancel.

Worked Examples:

Example 1: 1/2 × 3/4

No cross-cancel opportunity
Numerators: 1 × 3 = 3
Denominators: 2 × 4 = 8
Answer: 3/8

Example 2: 2/3 × 4/5

No common factors to cancel
Numerators: 2 × 4 = 8
Denominators: 3 × 5 = 15
Answer: 8/15

Example 3: 3/7 × 2/3

Cross-cancel: 3 in numerator and 3 in denominator
Becomes: 1/7 × 2/1 = 2/7
Answer: 2/7

Example 4: 5/6 × 3/10

Cross-cancel: 5 and 10 (÷5), 3 and 6 (÷3)
Becomes: 1/2 × 1/2 = 1/4
Answer: 1/4

Example 5: 4/9 × 3/8

Cross-cancel: 4 and 8 (÷4), 3 and 9 (÷3)
Becomes: 1/3 × 1/2 = 1/6
Answer: 1/6

1.3 Multiplying Fractions — More Examples

Example 6: 7/8 × 2/5

No easy cross-cancel
7 × 2 = 14, 8 × 5 = 40
Simplify: GCF(14,40)=2 → 7/20
Answer: 7/20

Example 7: 1/3 × 9/10

Cross-cancel: 3 and 9 (÷3)
Becomes: 1/1 × 3/10 = 3/10
Answer: 3/10

Example 8: 5/12 × 4/5

Cross-cancel: 5 and 5 cancel, 4 and 12 (÷4)
Becomes: 1/3 × 1/1 = 1/3
Answer: 1/3

Example 9: 6/7 × 7/12

Cross-cancel: 7 cancels, 6 and 12 (÷6)
Becomes: 1/1 × 1/2 = 1/2
Answer: 1/2

Example 10: 2/9 × 3/4

Cross-cancel: 3 and 9 (÷3), 2 and 4 (÷2)
Becomes: 1/3 × 1/2 = 1/6
Answer: 1/6

1.4 Dividing Fractions

Rule — "Keep, Change, Flip" (KCF):
To divide fractions:
• KEEP the first fraction exactly as it is.
• CHANGE the division sign (÷) to multiplication (×).
• FLIP the second fraction (write its reciprocal).
Then multiply normally and simplify.

What is a Reciprocal?

The reciprocal of a fraction is that fraction "flipped" — the numerator and denominator switch places.

• Reciprocal of 3/4 → 4/3

• Reciprocal of 2/5 → 5/2

• Reciprocal of 7 (or 7/1) → 1/7

Worked Examples:

Example 1: 1/2 ÷ 1/4

KEEP: 1/2
CHANGE: ÷ becomes ×
FLIP: 1/4 → 4/1
Multiply: 1/2 × 4/1 = 4/2 = 2
Answer: 2

Example 2: 3/4 ÷ 1/2

KEEP 3/4, FLIP 1/2 → 2/1
3/4 × 2/1 = 6/4 = 3/2 = 1 1/2
Answer: 1 1/2

Example 3: 2/3 ÷ 4/5

KEEP 2/3, FLIP 4/5 → 5/4
2/3 × 5/4 = 10/12 = 5/6
Answer: 5/6

Example 4: 5/8 ÷ 3/4

KEEP 5/8, FLIP 3/4 → 4/3
5/8 × 4/3 = 20/24 = 5/6
Answer: 5/6

Example 5: 7/10 ÷ 2/5

KEEP 7/10, FLIP 2/5 → 5/2
7/10 × 5/2 = 35/20 = 7/4 = 1 3/4
Answer: 1 3/4

1.4 Dividing Fractions — More Examples

Example 6: 4/9 ÷ 2/3

KEEP 4/9, FLIP 2/3 → 3/2
4/9 × 3/2 = 12/18 = 2/3
Answer: 2/3

Example 7: 1/6 ÷ 1/3

KEEP 1/6, FLIP 1/3 → 3/1
1/6 × 3/1 = 3/6 = 1/2
Answer: 1/2

Example 8: 5/6 ÷ 5/12

KEEP 5/6, FLIP 5/12 → 12/5
5/6 × 12/5 = 60/30 = 2
Answer: 2

Example 9: 3/5 ÷ 9/10

KEEP 3/5, FLIP 9/10 → 10/9
3/5 × 10/9 = 30/45 = 2/3
Answer: 2/3

Example 10: 8/9 ÷ 4/3

KEEP 8/9, FLIP 4/3 → 3/4
8/9 × 3/4 = 24/36 = 2/3
Answer: 2/3

Section 1: Practice Problems

Solve each problem. Show all work in the space below each problem.

Adding Fractions:

1) 3/8 + 1/8 =

2) 2/5 + 3/10 =

3) 1/3 + 5/6 =

4) 4/7 + 2/7 =

5) 5/12 + 1/4 =

Subtracting Fractions:

6) $7/8 - 3/8 =$

7) $5/6 - 1/3 =$

8) $3/4 - 1/6 =$

9) $11/12 - 2/3 =$

10) $9/10 - 3/5 =$

Multiplying Fractions:

11) $2/3 \times 3/4 =$

12) $5/8 \times 4/5 =$

13) $7/9 \times 3/7 =$

14) $1/2 \times 6/11 =$

15) $4/5 \times 5/6 =$

Dividing Fractions:

16) $3/4 \div 1/2 =$

17) $5/6 \div 2/3 =$

18) $7/8 \div 7/16 =$

19) $2/9 \div 4/9 =$

20) $1/3 \div 5/6 =$

Section 1 Quick Review:

• Adding/Subtracting → Need common denominator (LCD)

• Multiplying → Top × Top, Bottom × Bottom (cross-cancel first!)

• Dividing → Keep, Change, Flip → then multiply

• ALWAYS simplify your final answer!

Section 2: Decimals & Percentages

Decimals and percentages are everywhere in daily life — from prices at the store, to batting averages in sports, to calculating tips and sales tax. In this section, you will learn to convert between forms and perform operations with confidence.

2.1 Understanding Decimals

A **decimal** is another way to write a fraction whose denominator is a power of 10 (10, 100, 1000, etc.). The decimal point separates the whole number from the fractional part.

Place Values:
Ones . Tenths (1/10) | Hundredths (1/100) | Thousandths (1/1000)

2.2 Converting Between Fractions and Decimals

Fraction → Decimal: Divide the numerator by the denominator.
Decimal → Fraction: Use the place value as the denominator, then simplify.

Worked Examples:

Example 1: Convert 1/4 to a decimal

Divide numerator by denominator: 1 ÷ 4
= 0.25
Answer: 0.25

Example 2: Convert 3/8 to a decimal

3 ÷ 8 = 0.375

Answer: 0.375

Example 3: Convert 0.6 to a fraction

0.6 = 6 tenths = 6/10
Simplify: GCF(6,10) = 2 → 3/5
Answer: 3/5

Example 4: Convert 0.75 to a fraction

0.75 = 75 hundredths = 75/100
GCF(75,100) = 25 → 3/4
Answer: 3/4

Example 5: Convert 2/5 to a decimal

2 ÷ 5 = 0.4
Answer: 0.4

Example 6: Convert 0.125 to a fraction

0.125 = 125/1000
GCF(125,1000) = 125 → 1/8
Answer: 1/8

Example 7: Convert 7/20 to a decimal

7 ÷ 20 = 0.35
Answer: 0.35

Example 8: Convert 0.32 to a fraction

0.32 = 32/100
GCF(32,100) = 4 → 8/25
Answer: 8/25

Example 9: Convert 5/8 to a decimal

5 ÷ 8 = 0.625
Answer: 0.625

Example 10: Convert 0.05 to a fraction

0.05 = 5/100
GCF(5,100) = 5 → 1/20
Answer: 1/20

2.3 Decimal Operations

Adding/Subtracting Decimals:
Line up the decimal points vertically. Fill missing places with zeros. Add or subtract column by column. Bring the decimal straight down.

Multiplying Decimals:
Multiply as whole numbers. Count total decimal places in both factors. Place decimal that many places from the right in your answer.

Dividing Decimals:
Move decimal in divisor to make it a whole number. Move decimal in dividend the same number of places. Divide normally.

Worked Examples — Operations:

Example 1: 4.56 + 2.78

Line up: 4.56 + 2.78
Add: 6 + 8 = 14 (write 4, carry 1)
5 + 7 + 1 = 13 (write 3, carry 1)
4 + 2 + 1 = 7
Answer: 7.34

Example 2: 10.5 – 3.72

Line up: 10.50 – 3.72
Borrow as needed
= 6.78
Answer: 6.78

Example 3: 2.4×3.5

As whole numbers: $24 \times 35 = 840$
Count places: $1 + 1 = 2$ decimal places
Place decimal: 8.40
Answer: 8.4

Example 4: $6.72 \div 2.1$

Move decimals: $67.2 \div 21$
$= 3.2$
Answer: 3.2

2.4 Percentages

A **percent** means "per hundred" or "out of 100." The symbol % means /100.

Converting:
• Decimal → Percent: Multiply by 100 (move decimal 2 places right). Add %.
• Percent → Decimal: Divide by 100 (move decimal 2 places left). Remove %.

Finding Percent of a Number:
Convert percent to decimal, then multiply by the number.

Worked Examples:

Example 1: Find 25% of 80

Convert: 25% = 0.25
Multiply: 0.25 × 80 = 20
Answer: 20

Example 2: Find 10% of 150

Convert: 10% = 0.10
Multiply: 0.10 × 150 = 15
(Shortcut: move decimal 1 place left!)
Answer: 15

Example 3: Find 50% of 64

Convert: 50% = 0.50
Multiply: 0.50 × 64 = 32
(Shortcut: 50% = half!)

Answer: 32

Example 4: Find 15% of 200

Convert: 15% = 0.15
Multiply: $0.15 \times 200 = 30$
Answer: 30

Example 5: Find 75% of 120

Convert: 75% = 0.75
Multiply: $0.75 \times 120 = 90$
Answer: 90

2.4 Percentages — More Examples

Example 6: Find 8% of 50

Convert: 8% = 0.08
Multiply: $0.08 \times 50 = 4$
Answer: 4

Example 7: Find 33% of 90

Convert: 33% = 0.33
Multiply: $0.33 \times 90 = 29.7$
Answer: 29.7

Example 8: Find 60% of 45

Convert: 60% = 0.60
Multiply: $0.60 \times 45 = 27$
Answer: 27

Example 9: Find 5% of 300

Convert: 5% = 0.05
Multiply: $0.05 \times 300 = 15$
Answer: 15

Example 10: Find 120% of 50

Convert: 120% = 1.20 (more than 100%!)
Multiply: $1.20 \times 50 = 60$
Answer: 60

Section 2: Practice Problems

Solve each problem. Show your work.

1) Convert 3/5 to a decimal.

2) Convert 0.875 to a fraction in simplest form.

3) 4.56 + 2.78 =

4) 10.5 – 3.72 =

5) 2.4 × 3.5 =

6) 6.72 ÷ 2.1 =

7) Convert 0.45 to a percent.

8) Convert 72% to a decimal.

9) Find 30% of 90.

10) Find 15% of 240.

Section 2: Practice Problems (continued)

11) What percent is 12 of 48?

12) $0.008 \times 1000 =$

13) Convert 7/8 to a percent.

14) 25% of what number is 18?

15) $3.14 + 2.006 =$

16) $8.1 \times 0.03 =$

17) Convert 135% to a decimal.

18) Find 6.5% of 400.

19) $12.5 \div 0.25 =$

20) What is 80% of 55?

Section 3: Ratios & Proportions

Ratios and proportions help us compare quantities and solve real-world problems — from cooking recipes to map reading to calculating speed. These skills are essential for algebra and standardized tests like the LEAP.

3.1 Understanding Ratios

A ratio compares two quantities. It can be written three ways:
• As a fraction: 3/4
• With a colon: 3:4
• In words: "3 to 4"

Simplifying: Divide both parts by their GCF, just like fractions.

3.2 Unit Rates

Rule: A unit rate is a ratio where the second quantity is 1. To find it, divide both quantities by the second number.
Example: 150 miles in 3 hours → 150÷3 = 50 miles per 1 hour = 50 mph

3.3 Solving Proportions

Rule: A proportion says two ratios are equal: a/b = c/d
To solve: Cross-multiply → a × d = b × c
Then solve for the unknown variable.

Worked Examples:

Example 1: A bag has 6 red and 9 blue marbles. Ratio of red to blue?

Ratio = 6:9
GCF(6,9) = 3
Simplify: 6÷3 : 9÷3 = 2:3
Answer: 2:3

Example 2: Drive 150 miles in 3 hours. Find the unit rate.

Rate = 150 miles ÷ 3 hours
= 50 miles per hour
Answer: 50 mph

Example 3: Solve: 4/5 = x/20

Cross-multiply: 4 × 20 = 5 × x
80 = 5x
Divide: x = 80 ÷ 5 = 16
Answer: x = 16

Example 4: $12 for 4 pounds. Price per pound?

Unit rate = $12 ÷ 4
= $3 per pound
Answer: $3.00/lb

Example 5: Solve: 3/7 = 12/x

Cross-multiply: 3 × x = 7 × 12
3x = 84
x = 84 ÷ 3 = 28
Answer: x = 28

Section 3: More Examples

Example 6: 240 words in 4 minutes. Words per minute?

Unit rate = 240 ÷ 4
= 60 words per minute
Answer: 60 wpm

Example 7: Solve: $x/6 = 15/18$

Cross-multiply: $18x = 6 \times 15$
$18x = 90$
$x = 90 \div 18 = 5$
Answer: $x = 5$

Example 8: Recipe: 2 cups flour for 3 dozen cookies. How much for 9 dozen?

Proportion: $2/3 = x/9$
Cross-multiply: $2 \times 9 = 3x$
$18 = 3x \rightarrow x = 6$
Answer: 6 cups

Example 9: Solve: $5/8 = 30/x$

Cross-multiply: $5x = 8 \times 30$
$5x = 240$
$x = 48$
Answer: $x = 48$

Example 10: Map: 1 inch = 25 miles. Cities are 3.5 inches apart.

Distance = 3.5×25
= 87.5 miles
Answer: 87.5 miles

Section 3: Practice Problems

Solve each. Show your work.

1) Write the ratio 15 to 25 in simplest form.

2) A car travels 280 miles on 8 gallons. Find mpg.

3) Solve: 6/x = 18/27

4) Solve: x/12 = 5/4

5) 5 notebooks cost $8. How much for 15 notebooks?

6) Simplify 48:36.

7) Solve: 7/10 = x/50

8) A jogger runs 3 miles in 24 min. Minutes per mile?

9) Solve: 9/x = 3/7

10) 4 shirts cost $52. How much for 7 shirts?

Section 3: Practice (continued)

11) Simplify 100:75.

12) Solve: 2/5 = 14/x

13) Factory makes 360 items in 6 hours. Items/hour?

14) Solve: x/9 = 20/36

15) Map: 2 cm = 50 km. Distance for 7 cm?

16) Solve: 8/12 = x/15

17) Recipe: 3 eggs for 2 cakes. Eggs for 5 cakes?

18) Solve: 11/x = 33/12

19) 45 students : 3 teachers. Students per teacher?

20) Solve: x/8 = 45/72

Section 4: Integers & Absolute Value

Integers include all positive whole numbers, negative whole numbers, and zero. Understanding integers is crucial for algebra, science (temperature, elevation), and real-world math (bank accounts, gains and losses).

4.1 The Number Line

$\leftarrow\ -5\ \ -4\ \ -3\ \ -2\ \ -1\ \ 0\ \ +1\ \ +2\ \ +3\ \ +4\ \ +5\ \rightarrow$

• Numbers **increase** moving RIGHT. Numbers **decrease** moving LEFT.
• Zero is neither positive nor negative.
• Every positive number is greater than every negative number.

4.2 Absolute Value

Definition: The absolute value $|n|$ is the DISTANCE from zero on the number line. Distance is always positive (or zero).
$|5| = 5$, $|-5| = 5$, $|0| = 0$

4.3 Integer Operations Rules

Adding Integers:
• Same signs → Add absolute values, keep the sign.
• Different signs → Subtract absolute values, keep sign of the LARGER absolute value.

Subtracting Integers:
Change to addition of the opposite: $a - b = a + (-b)$. Then use addition rules.

Multiplying/Dividing:

- **Same signs → POSITIVE answer**
- **Different signs → NEGATIVE answer**

Worked Examples:

Example 1: $(-5) + (-3)$

Same signs (both negative)
Add absolute values: $5 + 3 = 8$
Keep the negative sign
Answer: –8

Example 2: $7 + (-4)$

Different signs
Subtract: $|7| - |-4| = 7 - 4 = 3$
Larger absolute value (7) is positive
Answer: 3

Example 3: $(-9) + 6$

Different signs
Subtract: $|-9| - |6| = 9 - 6 = 3$
Larger absolute value (9) is negative
Answer: –3

Example 4: $5 - 8$

Rewrite: $5 + (-8)$
Different signs: $8 - 5 = 3$
Negative wins ($8 > 5$)
Answer: –3

Example 5: $(-3) - (-7)$

Rewrite: $-3 + (+7) = -3 + 7$
Different signs: $7 - 3 = 4$
Positive wins
Answer: 4

Example 6: (–6) × 4

Different signs → negative
6 × 4 = 24
Answer: –**24**

Example 7: (–8) × (–5)

Same signs → positive
8 × 5 = 40
Answer: 40

Example 8: (–36) ÷ 9

Different signs → negative
36 ÷ 9 = 4
Answer: –**4**

Example 9: |–12|

Distance from zero = 12
Answer: 12

Example 10: |7| + |–3|

= 7 + 3
Answer: 10

Section 4: Practice Problems

Solve each. Show work.

1) (–7) + 3 =

2) 5 + (–9) =

3) (–4) + (–6) =

4) 8 – 12 =

5) (–3) – 5 =

6) (–10) – (–4) =

7) 6 × (–7) =

8) (–9) × (–3) =

9) (–48) ÷ 8 =

10) 56 ÷ (–7) =

Section 4: Practice (continued)

11) $|-15| =$

12) $|8 - 13| =$

13) $(-2) \times 5 \times (-3) =$

14) $(-1) + (-1) + (-1) =$

15) $0 - (-9) =$

16) $|-4| + |-7| =$

17) $(-3)^2 =$

18) $-3^2 =$

19) Order: 3, –5, 0, –2, 7 (least to greatest)

20) Which is greater: –8 or –3? Explain.

Section 5: Expressions & Equations

Algebra begins with expressions and equations. An expression is a math phrase with numbers, variables, and operations. An equation is a statement that two expressions are equal. Learning these skills is your gateway to high school math success.

5.1 Writing Algebraic Expressions

Key Translation Phrases:

• "sum" / "more than" / "increased by" → + (addition)

• "difference" / "less than" / "decreased by" → – (subtraction)

• "product" / "times" / "of" → × (multiplication)

• "quotient" / "divided by" / "per" → ÷ (division)

5.2 Evaluating Expressions

Rule: Substitute the given value for the variable, then simplify using PEMDAS (Parentheses, Exponents, Multiply/Divide, Add/Subtract).

5.3 Solving One-Step Equations

Rule: Use inverse operations to get the variable ALONE on one side.

• If the equation has + → subtract from both sides

• If the equation has – → add to both sides

• If the equation has × → divide both sides

• If the equation has ÷ → multiply both sides

Worked Examples:

Example 1: Write: 5 more than a number n

"more than" = addition

Answer: $n + 5$

Example 2: Write: 3 times a number decreased by 7

"3 times a number" = $3n$

"decreased by 7" = $– 7$

Answer: $3n – 7$

Example 3: Evaluate $2x + 5$ when $x = 4$

Substitute: $2(4) + 5$

$= 8 + 5 = 13$

Answer: 13

Example 4: Evaluate $3a^2 – 1$ when $a = 3$

Substitute: $3(3^2) – 1$

$= 3(9) – 1 = 27 – 1 = 26$

Answer: 26

Example 5: Solve: $x + 7 = 15$

Inverse of $+7$ is $–7$

$x + 7 – 7 = 15 – 7$

$x = 8$

Answer: $x = 8$

Example 6: Solve: $y – 4 = 12$

Inverse of $–4$ is $+4$

$y – 4 + 4 = 12 + 4$

y = 16

Answer: y = 16

Example 7: Solve: 3m = 24

Inverse of ×3 is ÷3

3m ÷ 3 = 24 ÷ 3

m = 8

Answer: m = 8

Example 8: Solve: n/5 = 6

Inverse of ÷5 is ×5

n/5 × 5 = 6 × 5

n = 30

Answer: n = 30

Example 9: Solve: x + 2.5 = 10

Subtract 2.5: x = 10 – 2.5

Answer: x = 7.5

Example 10: Solve: 4w = –20

Divide by 4: w = –20 ÷ 4

Answer: w = –5

Section 5: Practice Problems

Show all work.

1) Write: twice a number plus 9.

2) Write: a number divided by 4, minus 3.

3) Evaluate $5x – 2$ when $x = 6$.

4) Evaluate $2(a + 3)$ when $a = 5$.

5) Evaluate $x^2 + 4x$ when $x = 3$.

6) Solve: $x + 9 = 21$

7) Solve: $y – 6 = 14$

8) Solve: $5n = 45$

9) Solve: $m/3 = 12$

10) Solve: $x + 15 = 8$

Section 5: Practice (continued)

11) Solve: $2x - 4 = 10$ (two-step)

12) Write: the product of 7 and a number, increased by 2.

13) Evaluate $4a - b$ when $a = 3$ and $b = 5$.

14) Solve: $w/6 = -3$

15) Solve: $x - 2.4 = 5.6$

16) Solve: $8k = 72$

17) Evaluate $|2x - 10|$ when $x = 3$.

18) Solve: $n + (-4) = 7$

19) Solve: $-3x = 27$

20) Write and solve: A number plus 12 equals 5.

Section 6: LEAP 2025 Practice Test

40 Mixed Problems — Standardized Test Format

Directions: Read each problem carefully. Show ALL of your work in the space provided. Write your final answer clearly. No calculators allowed.

Tips: Check your work if you finish early. Estimate first to check if your answer is reasonable. Don't leave any questions blank!

1) Simplify: $3/4 + 5/8$

2) What is 35% of 160?

3) Solve for x: $x/4 = 12$

4) A shirt costs $45. It is on sale for 20% off. What is the sale price?

5) Evaluate $3x^2 - 2x + 1$ when $x = 2$.

6) $(-8) + 5 - (-3) =$

7) Convert 0.375 to a fraction in simplest form.

8) Solve: $2/3 = x/15$

LEAP 2025 Practice Test (continued)

9) A rectangle has length 12 cm and width 8 cm. Find its perimeter.

10) What is the absolute value of –17?

11) Simplify: 5/6 × 3/10

12) A car travels 210 miles in 3.5 hours. What is the speed in mph?

13) Solve: n – 7 = –3

14) Convert 7/20 to a decimal and a percent.

15) Find the unit rate: $6.48 for 4 pounds.

16) (–4) × (–9) =

LEAP 2025 Practice Test (continued)

17) Simplify: $7/8 \div 1/4$

18) Write an expression: 6 less than twice a number.

19) Solve: $5x = -35$

20) Which is larger: 3/5 or 7/12? Show your reasoning.

21) Map scale: 1 inch = 40 miles. Cities are 3.5 in. apart. Actual distance?

22) Solve: $x + 4.5 = 11.2$

23) Convert 125% to a decimal.

24) $(-6)^2 =$

LEAP 2025 Practice Test (continued)

25) Simplify: 2/3 – 1/4

26) A bag has 4 red, 6 blue, 5 green marbles. Ratio of red to total?

27) Solve: m/8 = 7

28) Find 8% of 250.

29) Order from least to greatest: –4, 2, –7, 0, 5

30) 3.6 × 2.5 =

31) Solve the proportion: 9/12 = x/20

32) Write the prime factorization of 72.

LEAP 2025 Practice Test (continued)

33) $(-14) \div (-2) =$

34) Evaluate $2(x + 3) - 4$ when $x = 5$.

35) Convert 3/8 to a decimal.

36) A recipe calls for 2/3 cup sugar. Triple the recipe — how much sugar?

37) Solve: $x - (-5) = 12$

38) What percent of 60 is 45?

39) Simplify: $|(-3) + (-8)|$

40) Student scores: 85, 92, 78, 95. Find the average.

LEAP 2025 — Complete Answer Key

1) 11/8 = 1 3/8
2) 56
3) x = 48
4) $36.00
5) 9
6) 0
7) 3/8
8) x = 10
9) 40 cm
10) 17
11) 1/4
12) 60 mph
13) n = 4
14) 0.35 = 35%
15) $1.62/lb
16) 36
17) 7/2 = 3 1/2
18) 2n – 6
19) x = –7
20) 3/5 > 7/12
21) 140 miles
22) x = 6.7
23) 1.25
24) 36
25) 5/12
26) 4:15
27) m = 56
28) 20
29) –7, –4, 0, 2, 5

30) 9.0

31) $x = 15$

32) $2^3 \times 3^2$

33) 7

34) 12

35) 0.375

36) 2 cups

37) $x = 7$

38) 75%

39) 11

40) 87.5

Course Summary — Key Formulas & Rules

FRACTIONS:

- Add/Subtract: Find LCD → rewrite → add/subtract numerators → simplify
- Multiply: Top × Top, Bottom × Bottom (cross-cancel first)
- Divide: Keep, Change, Flip → then multiply

DECIMALS:

- Add/Subtract: Line up decimal points, fill zeros
- Multiply: Multiply as whole numbers, count decimal places
- Divide: Make divisor a whole number, move both decimals same way

PERCENTAGES:

- Decimal → %: Multiply by 100
- % → Decimal: Divide by 100
- Percent of a number: Convert to decimal × number

RATIOS & PROPORTIONS:

- Simplify ratios like fractions (divide by GCF)
- Unit rate: Divide to get denominator = 1
- Solve proportions: Cross-multiply → solve

INTEGERS:

- Same signs: Add, keep sign
- Different signs: Subtract, keep sign of larger |value|

• Multiply/Divide: Same signs → +, Different signs → –

EQUATIONS:

• Use inverse operations to isolate the variable

• Whatever you do to one side, do to the other

• PEMDAS for order of operations

Congratulations on completing this course! Keep practicing and you will master these skills.

"Dreams can come true. Better late than never."

— Gregory Lazard Jr.